

Polymer informatics: Expert-in-the-loop in QSPR modeling of refractive index

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ABSTRACT

Refractive index (RI) is a highly relevant property for the design of new polymeric materials for very specific applications in the telecommunications industry, medicine, and analytical chemistry, among many others. A particular case is that of plastic optical fibers, in which the information is transmitted by photons and then RI takes center stage. Therefore, the modeling and prediction of this property play a key role when characterizing and designing materials for these important industries. Over the last decades, the use of Machine Learning (ML) algorithms in the modeling of properties for the design of new materials has been consolidated thanks to the gradual increase in the available databases. In particular, the development of Quantitative Structure-Property Relationship (QSPR) models has benefited from these emerging technologies, providing the possibility of generating *in silico* testing strategies applicable to the early stages of the design of new materials. However, in many cases, it has been observed that using ML algorithms in a fully automatic way, without human intervention in the QSPR model design process, tend to generate black-box models that have a difficult interpretation and can lose sight about relevant aspects that require both criteria and an expert's knowledge in the chemical domain. For this reason, interactive ML methodologies that combine computational outputs with experts' knowledge, usually known as expert-in-the-loop strategies, are becoming more frequent. In this article, we present the design of QSPR models for RI modeling following two different approaches, a black-box ML methodology and an Interactive Machine Learning (IML) methodology with expert-in-the-loop, from a database whose curation is also described in the present work. In this regard, visual analytics strategies were used to capture the expert's knowledge, facilitating an effective and rapid interaction between the outputs provided by ML and the chemical analyst. In addition, we contrast the best models obtained by both approaches against two other predictive models for RI estimation reported in the literature, achieving promising performances in terms of cardinality and accuracy when the expert interacts during modeling. In summary, the obtained results allow us to claim that the expert-in-the-loop approach provides QSPR models with better generalizability properties and more interpretable from a physicochemical point of view, without losing accuracy. Finally, in addition to providing high quality QSPR models to predict the RI of polymeric materials, the present work lays the foundation for defining an effective methodology to incorporate experts' knowledge in the design of other material properties.

1. Introduction

In today's world, communications play a major role, and technology aims to develop solutions to increase their efficiency and position competitive mass-manufactured products. A particular case is that of optical fibers, which have traditionally been manufactured with a SiO₂

core, and for more than four decades have also been manufactured by combining polymers, the so-called plastic optical fibers (POFs). This kind of fibers are not only used in telecommunications but also in medical, analytical chemistry, aerospace, and industrial applications in general, among many others. As regard the advantages of POFs, they are lightweight, flexible (the traditional optical fiber is rigid), and low cost;

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handling and maintenance costs are potentially low as well. In this sense, the demand for polymeric materials that cover the needs of these products has increased exponentially, due to their versatility and the growing use for domestic installations. Nevertheless, the data transmission speed of POFs still presents challenges, since it does not reach the one of SiO₂ fibers [1–3]. To develop new polymeric materials, a set of properties related to their flexibility, strength, adhesion, and interaction with light must be considered. This last aspect is contemplated in the optical properties, which take the leading role because the data transmission phenomenon is carried out through electromagnetic waves (photons). Therefore, it is of great interest to know what optical characteristics, in particular refractive index (RI), a material must have to transmit data efficiently. Optical fibers are formed by a core and several layers, but the most important elements to consider are the core and the immediate layer that covers it (cladding). They must have a difference in RI that makes the light follow the path along the core, without deviation; this is achieved with a core with higher RI than the one of the cladding. Most existing synthetic polymers have an RI of about 1.5 that covers a range from 1.3 to 1.7. In addition, there is now a growing interest in high RI polymers as certain applications, such as lenses, anti-reflective coatings, and image sensors, require less material the higher the RI. [4,5]. Therefore, due to the increasing demand for materials with specific characteristics and low cost, it becomes necessary to create them from the molecular design stage.

A polymeric material designer's work is dynamic and interdisciplinary and, in the early stages, uses increasingly available computer tools. The most required ones are those that can predict properties from the structural representation of the new designed polymer, since they quickly provide an estimate of the property that can guide the expert before synthesis. In this sense, the Quantitative Structure-Property Relationship (QSPR) methodology has had great advances, providing contributions for multiple properties of polymeric materials, as well as for RI [6–9]. The first contributions on these topics are found in the literature by Bicerano [10], whose database (DB) contains 183 polymers and has been used in numerous subsequent works. Bicerano [10] reported a model with a cardinality of 10 descriptors and an R^2 of 0.954. Below are mentioned only some relevant works on this topic that will serve as a basis for comparison with ours. Xu et al. [11] worked with a DB of 121 polymers and obtained a model with an R^2 of 0.929 with four descriptors obtained by calculation. In a later work, Xu et al. [12] developed QSPR models based on a DB containing 120 polymers, achieving an R^2 of 0.943 with a linear model and an R^2 of 0.971 with a non-linear model, both of five descriptors, using 2D and 3D descriptors, among other classes of descriptors. Astray et al. [13] used 95 polymers to train an Artificial Neural Network achieving an R^2 of 0.870. Duchowicz et al. [14] worked with a DB of 234 polymers and obtained a single flexible descriptor model achieving an R^2 of 0.96. Jabeen et al. [15] presented a model of four descriptors, achieving an R^2 of 0.932. In addition, they not only presented the model, but also created virtual libraries that they used to predict property values. Khan et al. [16] did QSPR modeling using 2D descriptors. The DB they used consists of 221 polymers, developing four models of six descriptors each, whose best performance is with an R^2 of 0.899. This last work also provides a detailed review of how the prediction of RI is approached [16]. Nevertheless, this topic continues to be of great interest and presents several challenges; recently Lightstone et al. [17] used ML to model this property with a database of more than 500 polymers of different chemical families, achieving an R^2 of 0.880. Note that the QSPR methodology is databased; therefore, it is supported by Machine Learning (ML) techniques that facilitate and accelerate the resolution of complex, and otherwise intractable, problems. However, the use of some popular ML algorithms applied to QSPR modeling, such as Support Vector Machine and Neural Networks, as an automatic way, without the chemical analysts' intervention in the QSPR model design process, usually generates black-box models that have a difficult physicochemical interpretation. Moreover, one of the most outstanding challenges in materials science is

that relatively small experimental databases are used in contrast to the chemical space in which the new materials to be predicted could be found. In this sense, introducing chemists' expertise in the field through Interactive Machine Learning (IML) to train models with relevant features and with a physicochemical sense related to the target property to achieve generalizable predictive models is one of the current topics in QSPR modeling [18–21].

In this paper, which is part of the line of polymer informatics of our research group [22–25], we present the results of QSPR modeling to predict the RI property of synthetic polymers. As an original contribution, two QSPR methodologies are presented in a comparative way: a black-box ML approach and an IML approach with expert-in-the-loop. The aim was to evaluate strategies with a polymeric material expert's intervention. In this sense, we intended to answer the following question: *When experts intervene by providing knowledge in the different stages of modeling, can they effectively contribute to reduce the cardinality and improve the interpretability of QSPR models while preserving their levels of precision?*

2. Materials and methods

This section starts with the description of the target property, the construction of the DB, and the calculation of the descriptors. Then, following a methodological scheme for a better understanding, the two approaches followed for the RI QSPR modeling are explained: the black-box approach and the expert-in-the-loop approach (Fig. 1).

2.1. Target property and database

2.1.1. Refractive index

Refractive index is the most important optical property of polymers. The speed of light passing through a polymer is affected by the polarity of the bonds in the molecule; this is associated with the dielectric constant of the material. As a result, the light beam changes direction when it passes from one medium to another. Although the chemical structure and morphology of the material affect optical properties in general, in the present work we focus on the physical aspects underlying the RI in particular. Specifically, we focus on polarity/polarizability, which can be studied from three aspects: orientational, atomic, and electronic. These aspects are defined by the orientation of the constant moments in a field, the relative displacement of the positive and negative ions inside the molecule, and the displacement of the electrons in relation to the nuclei [26]. Therefore, when working on QSPR modeling, it is important to recognize the molecular descriptors that capture these aspects that influence the value of the property.

2.2. Database

In the context of working with ML, it is always desirable to have DBs with as many molecules (instances) as possible. For this reason, the DB used in this work is the product of a thickening of a previous base cured in an initial work on RI [27]. The process followed to obtain the 227 polymers that conform the new clean and cured DB is detailed below:

- I. One hundred and twenty-six polymers were taken from the DB cleaned and cured by us following the methodology explained in Schustik et al. [27]. The DB in this last work had been obtained from Jabeen et al. [15]. These 126 polymers are represented by the corresponding structural repetitive unit (SRU) in SMILES code.
- II. One hundred and one polymers were added to the 126 polymers mentioned in I; they were taken from the DB provided by Duchowicz et al. [14], which is composed of 234 polymers. The process began with a control and check of both the SRU structure and the corresponding property value in each of the 234 molecules. The molecular representation used was SMILES code. Several sources were followed as reference to carry out this control [10,11,28–30]. During the

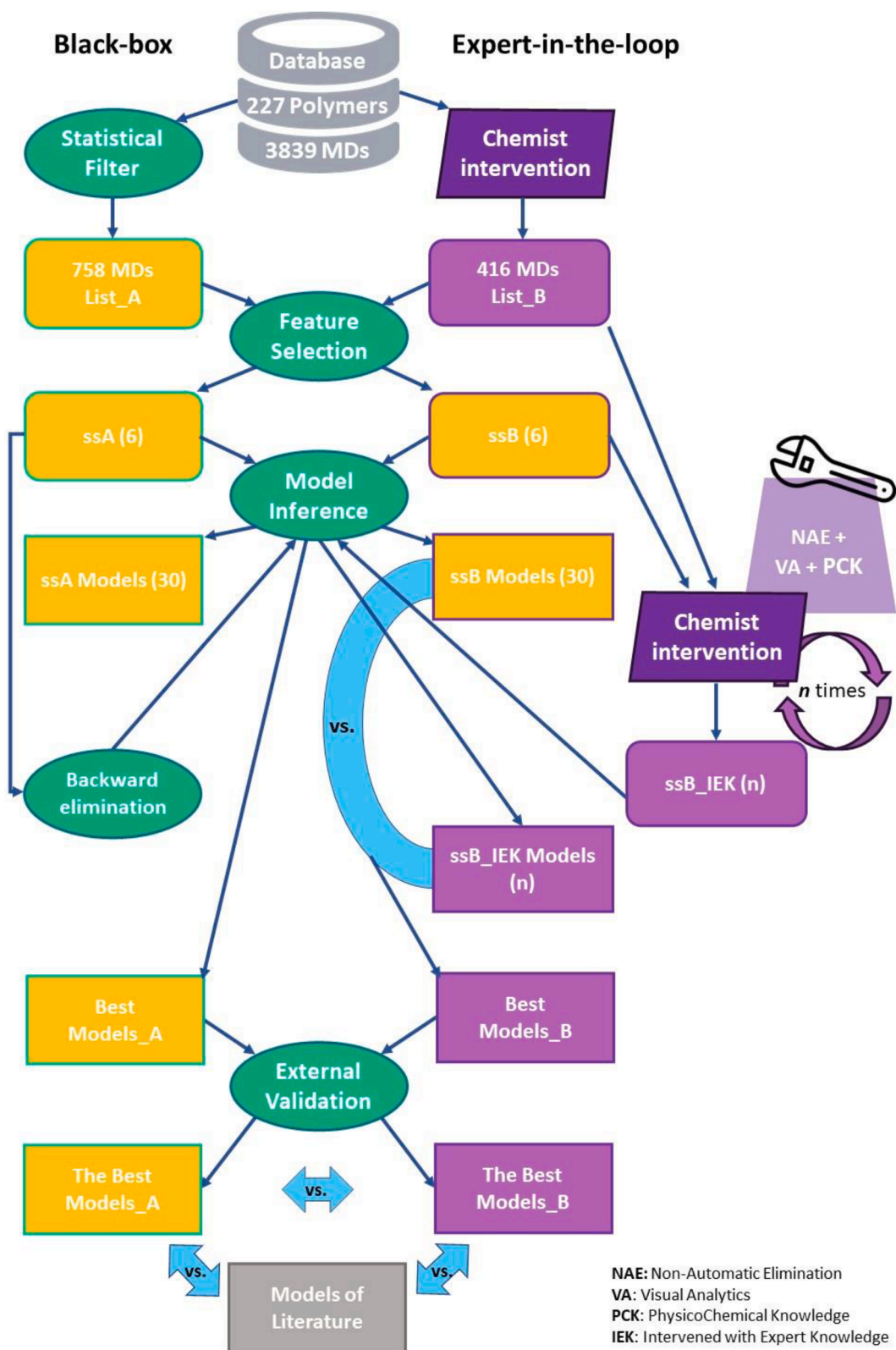


Fig. 1. Methodological scheme comparing the black-box approach and the expert-in-the-loop approach.

curing process, some duplicated polymers and errors in the structure and property value were found and corrected as follows:

A. In total, 119 polymers were removed; some of them were duplicates in the DB by Duchowicz, and others were common with our initial database (I).

B. Errors in 12 molecules were corrected against the reference sources.

C. Some polymers that could not be validated because they were not found in the reference sources were discarded. In addition, others whose property values presented inconsistencies between DBs were also discarded. In total, 14 polymers were eliminated.

Thus, as a result of this process, 101 cured polymers were added to the initial DB, making a total of 227 final polymers, representing an increase of 80%. All the collected property values were measured at 298 K. The DB is presented in [Table S.1 of Supplementary material](#). The vast majority of polymers in this database are amorphous. For those that are semi-crystalline such as polyethylene, the RI value was taken in the limit of 0% crystallinity [10]. Different families make up the database: polyolefins, polyoxides, polyethers, polyesters, polyacetates, polyvinyls, polyamides, polyimides, polyketones, polyphenylenes, polycarbonates, polypthalates, polymethacrylates, polyacrylates, polysilylenes, polysulfides, and polysulfones. The applicability domain of the QSPR models of this work corresponds to the chemical space characterized by these families.

2.3. Molecular descriptors

2.3.1. Classic molecular descriptors: List_A

Classical molecular descriptors are numerical values that provide physicochemical information calculated from the computational representation of the molecule. For our DB, we used the representation called Simplified Molecular Input Line Entry Specification (SMILES), which allows obtaining a compact 2D representation of the molecular structure for each polymer. From these representations, 3839 DMs were obtained for each of the 227 polymers using the DRAGON tool [31]. A 95% correlation filter between descriptors was applied, and those with at least one non-assigned (N/A) value were eliminated. In this way, 758 descriptors were obtained, which we named as List_A.

2.3.2. Polar molecular descriptors: List_B

In addition to List_A, a list called List_B was constructed with only those descriptors that somehow possess information on the polar aspect of the polymer molecule (See [Section 2.1](#) Target property). Therefore, not only those that obviously provide polarity or polarizability information, such as polar group descriptors and aromaticity descriptors, but also those that contain solubility information, such as ALOGP, were chosen with expert criteria, since a substance has a different solubility according to its polarity. This new list was constructed in two stages. First, from the total universe of descriptors computed by DRAGON (4885), 612 were chosen that satisfied the expert's selection criteria. Second, those with non-assigned (N/A) values and those that Dragon cannot calculate for the SMILES type representation were eliminated. Thus, List_B was made up of 416 descriptors. At this point, the expert decided not to filter the pre-selected descriptors linearly correlated with each other during this first stage, given that sometimes it is useful to have redundant descriptors in order to achieve final models with high interpretability. This is due to the fact that redundant descriptors represent variables that provide the same information to the model and, therefore, among these alternatives of similar descriptors you can choose those that have a better interpretability in physicochemical terms for the model. In brief, the redundancy during initial steps of the analysis provides alternatives to the expert. Obviously, in the final selection of descriptors that will make up the final QSPR model, there should be no redundant variables. Therefore, List_B is not a subset of List_A. Later, during the visual analysis, those descriptors that provide redundant information and are complex to interpret in physicochemical terms can be detected and eliminated.

2.4. Feature selection stage

Feature selection (FS) is an important stage in an ML process. FS is a dimensionality reduction methodology and is used to reduce the number of features (molecular descriptors) in a dataset. This dimension reduction is not done only to reduce the computational cost, but mainly to reduce the effect of noise and to extract the main features from the data.

The training set is used for the FS stage. In the present work, two methodological approaches were followed. One of them is exclusively

based on the automatic processes for FS and backward elimination (black-box approach) applied in List_A (set of 758 MDs). The other one adopts the chemist's intervention (expert-in-the-loop approach) improving the automatic FS, applied in List_B (set of 416 MDs), by visual analytics. This list is derived from the expert's knowledge, as explained in detail in the previous subsections. The WEKA tool [32] was employed for FS, using fixed seed randomization and default settings. The following FS methods were used: Correlation-based Feature Subset Selection (CFS) and WrapperSubsetEval. The last one was combined with Linear Regression (WLR), Neural Networks (WNN), Random Forest (WRF), Random Committee (WRC), and K-Nearest Neighbours (WKNN) to estimate the accuracy for each set of DMs (attributes). Finally, 12 subsets were obtained, six from List_A (ssA1 to ssA6) and six from List_B (ssB1 to ssB6). Following the black-box approach, the six subsets obtained from List_A, were called: ssA1 (selected by CFS), ssA2 (selected by WNN), ssA3 (selected by WRC), ssA4 (selected by WRF), ssA5 (selected by WLR), and ssA6 (selected by WKNN). In the methodological scheme ([Fig. 1](#)), they are shown as "ssA". Following the expert-in-the-loop approach, the six subsets obtained from List_B were called: ssB1 (selected by CFS), ssB2 (selected by WNN), ssB3 (selected by WRC), ssB4 (selected by WRF), ssB5 (selected by WLR), and ssB6 (selected by WKNN). In [Fig. 1](#), they are shown as "ssB".

2.5. QSPR models inference stage

In this stage, the training set was used; the ML methods applied to train the QSPR models were: Linear Regression (LR), Neural Networks (NN), Random Forest (RF), Random Committee (RC), and K-Nearest Neighbours (KNN) provided by the WEKA tool [32]. The WEKA parameters were used by default, and the leave-one-out-cross-validation (LOOCV) was used as a model validation technique to test each model. Our decision to use LOOCV aimed at maintaining a fair comparison with the two models published in the literature against which we compared. In this regard, the models reported by Jabeen [15] and Khan [16] use LOOCV. The coefficient of determination (R^2) was the metric considered to evaluate the performance of QSPR models.

Following the methodological scheme, it can be seen at this stage that each approach obtained 30 models as output, both products of the six subsets combined with the five ML methods, "Models_A" and "Models_B", respectively (dark yellow boxes in [Fig. 1](#)). From this point on, it is convenient to explain each approach separately, since the expert's intervention was intensified to explore our hypothesis, in which feature engineering came to the fore [33,34].

2.5.1. Black-box approach

The steps involved in this approach were relatively simple. A backward elimination strategy was used to improve or maintain the accuracies of the models. The backward elimination is a well-known method and it was chosen because of its simplicity, since a human expert should be able to imitate the process to make a fair comparison between methodologies. The two best models were chosen out of the 30 obtained based on R^2 and reducing their cardinality without losing performance. From this process, two models were also obtained, which were the reduced representatives of the previous ones and later passed to the external validation stage.

2.5.2. Expert-in-the-loop approach

To reduce the cardinality of the subsets from ssB, we used a strategy that combines the non-automatic elimination process, visual analytics, and an expert's physicochemical knowledge. Note that this strategy also needs the computation of metrics from QSPR models to improve or maintain the accuracies of the models. The ssBs were treated with visual analytics, using the VIDEAN tool [35]. Among the options offered by VIDEAN, there are graphs that show the correlation between pairs of descriptors in terms of Spearman, Kendall, and also based on entropy. VIDEAN also offers the visualization of the co-occurrence of MDs

considering all the ssBs. This means that some pairs of descriptors may appear together in more than one subset out of the six, and this could imply that they hold complementary information. Finally, VIDEAN offers the possibility of obtaining scatter plots of the behavior of the descriptors against the property; by comparing them among several MDs, it can be very easily seen how the information is distributed in the plot space. Thanks to these options and the possibility of performing interactive analyses in a simple and fast way, experts can guide a new selection of MDs with the aim of decreasing the cardinality and increasing the performance of the models taking advantage of their knowledge of the property and the study materials through an IML process. The tool can be freely accessed through the web page: <http://lidecc.cs.uns.edu.ar/VIDEAN/>.

Methodologically, in all the stages carried out, the QSPR models were inferred from each built subset to make the corresponding comparison using the R^2 value. There was an initial step of identification of the prevalent MDs, which were those selected in more than one subset. They were gathered in a new subset named ssBPrev. Thus, two strategies used by the expert to reduce the MDs can be explained: non-automatic elimination and visual analytics.

1. Non-automatic elimination: It consisted of eliminating one by one the MDs of the subset under analysis so as to always maintain the original cardinality minus 1 (i.e., each MD that was removed was then returned, and then another one was removed). Next, the MDs that, when removed, generated a decrease in the initial R^2 were identified, because it was deduced that these MDs were relevant to the property, thus obtaining the so-called relevant MDs.
2. Visual analytics: The information of the six ssBs plus the ssBPrev was loaded in VIDEAN. We started from the pair appearance graph (co-occurrence) to detect in a single step those MDs that appear in at least three models (dark green links). They were also considered relevant. Then, the analysis of the behavior of these MDs with the property was carried out with scatter plots and the correlation graph.
3. These strategies should be applied in combination with physico-chemical knowledge. With the relevant MDs obtained using one strategy or the other (1 or 2), experts decide, depending on the case, to add information that they consider to be missing from the physicochemical point of view

In the present work, the performances of the best models obtained from these reduced cardinality subsets were compared, and those that followed the selection criteria (high accuracy, low cardinality, and high interpretability) were chosen. Note that the expert was guided by the result of each step and could apply visual analytics as support also in the non-automatic elimination.

2.6. QSPR models validation stage

The generalizability of a model is evaluated using an external validation set, that is, a dataset that has not been involved in the whole QSPR model inference process. Three validation tests were considered in the present paper. The first one analyzes the R^2 values reached by the inferred models and the values of the external validation stage, for which the gap between them is calculated (Training - External validation). If the gap value does not exceed 0.05, it is suitable to advance to the next test. The second test analyzes the MAE values considering the sample evidence and the probability theory, since it consists of performing a hypothesis test whose H_0 is that the average of the training MAE is significantly higher than the average of the external validation MAE. These models for which H_0 can be rejected are considered free of overfitting; therefore, it is possible to consider them for the third validation test. This test also consists of analyzing the values of MAE and its standard deviation according to the algorithm proposed by Roy et al. [36], which allows to rule out the possibility of systematic errors and to classify the model prediction according to quality level (Good, Bad, or

Moderate).

Other options to evaluate the performance of a model are random experiments, such as the Y-Randomization test [37]. This technique allows evaluating the risk of correlation by chance, comparing the performance of the final model with the average performance of multiple models (1000 in this case) obtained using the same descriptors of the original model and random target values (permuted Y-data).

3. Results and discussions

3.1. Subsets analysis from feature selection

Feature selection was made from the two lists of descriptors separately. This resulted in six subsets of MDs for each list, expressed as ssA and ssB in Fig. 1. The cardinalities of all subsets can be seen in Table 1, while their detailed composition is found in Table S.2 in Supplementary material. In both the black-box and expert-in-the-loop approaches, the WLR method returns the subsets with the highest cardinality whereas the WRC method returns the subsets with the lowest ones. There is a rule of thumb that states that for each descriptor there should be at least five molecules in the DB [38]. This rule is fulfilled, because there are 170 molecules in the training set and the subsets of MDs are up to 26 descriptors. However, the cardinality of the subsets is high in all cases in comparison with those proposed in the literature to model RI, which are in the order of four MDs [11,12,15,16].

The selected MDs from List A are 80, and those from List B are 60. Note that 21 out of the 132 possible polar MDs were selected in List A; that is, 26% of the MDs automatically selected were identified as appropriate for predicting the target property. The prevalent MDs (i.e., those that are in more than one subset) of list A, are 11: **Mi**, Si-111, **GATS1p**, SpPosA_B(v), **nS**, SpMAD_B(m), **nN**, **GATS2p**, P_VSA_MR_8, SpMax_B(s), and B05[C-Si]. The prevalent MDs from List B are 17: **Mi**, **GATS2p**, F04[C-Cl], **GATS1e**, **GATS1p**, **N%**, **nS**, ALOGP, ATSC2e, Eta_betaP_A, MATS1p, **nAB**, **nConj**, **nCL**, **nF**, SM1_Dz(p), and SpMax1_Bh(p). Four prevalent MDs share both lists (highlighted in bold). A peculiarity is that the descriptor **Mi** (mean first ionization potential_scaled on Carbon atom) appears in all subsets of both lists, which denotes a load of information relevant to the prediction of the target property. The same was observed in Khan et al. [16], where each of the four reported models contained **Mi**.

3.2. Performance analysis of the inferred QSPR models

First, the five models explained in Section 2.4 for each of the six subsets were inferred through ML methods, obtaining 30 models for List A and other 30 models for List B, whose performances and values of the different errors for training are shown in Table S.3 in Supplementary material. For List A, the best model obtained was for ssA5 with LR as a QSPR inference method (ssA5_LR), achieving an R^2 of 0.9686, whereas the worst model was obtained for ssA3 with NN as a QSPR inference method (ssA3_NN), with an R^2 of 0.7903. For List B, the best model obtained was for ssB5 with LR as a QSPR inference method (ssB5_LR) and an R^2 of 0.9442; the worst model was obtained for ssB3 with NN as a QSPR inference method (ssB3_NN) and an R^2 of 0.8000. There was a

Table 1
Cardinality of the subsets from List A and List B.

FS method	Black-box approach (List A) Subset name: cardinality	Expert-in-the-loop approach (List B) Subset name: cardinality
CFS	ssA1: 9	ssB1: 12
WNN	ssA2: 21	ssB2: 13
WRC	ssA3: 8	ssB3: 10
WRF	ssA4: 17	ssB4: 15
WLR	ssA5: 26	ssB5: 25
WKNN	ssA6: 19	ssB6: 13
Prevalents	ssAPrev: 11	ssBPrev: 17

similar gap between the best and the worst models and a method coincidence with List_A. In both approaches, most models have a performance within the range of $0.80 \leq R^2 \leq 0.95$. So far, the procedure of this stage was the same in both approaches, but from now on each one has its own steps (See Fig. 1). Analyzing more deeply, although the performances were competitive, the cardinality, as mentioned above, was not ideal (Section 3.1). Therefore, neither model could be chosen as the best or the definitive one; strategies to reduce cardinality without losing predictive capacity had to be applied in both approaches.

On the one hand, following the black-box approach, the two best QSPR models were chosen as performance references, namely ssA5_LR (R^2 of 0.9686, #26) and ssA2_NN (R^2 of 0.9091, #21) (See Table S.3 of Supplementary material). The objective was to subject the subsets to a backward elimination reduction, where the only condition is to maintain or increase the performance of the model of the previous step. As a result, the cardinality of ssA5 was reduced by 20%, from 26 to 21 MDs, maintaining an R^2 of 0.9686 with LR (ssA5_#21a_LR). At the same time, ssA2 was reduced by 67%; that is, its cardinality decreased from 21 to 7 MDs, and the performance increased from an R^2 of 0.9091 to one of 0.9143 (ssA2_#7a_LR). These results, which correspond to the training, can be seen in Table 2.

On the other hand, in the expert-in-the-loop approach, the expert made multiple interventions from the six ssB, as well as from the ssBPrev (subset of prevalent MDs), following the criteria explained in Section 2.4. All the results for training are reported in Table S.3 in Supplementary material, and only the best performing models are shown in Table 2. While the models in List_A performed better, the models in List_B achieved competitive performances with a considerable reduction of cardinality.

To show the importance of the expert's contribution, some examples of how decisions were made to condense the information into models of low cardinality, high performance, and high physicochemical interpretability are given below. First, two types of analysis are shown, which were the beginning of the interventions: one based on elimination (non-automatic) and another based on visual analytics using the plots for mutual information and co-occurrence of pairs as well. Finally, with a third example, it is shown how the expert intervenes by adding MDs considered relevant to complete the information space.

3.2.1. Non-automatic elimination strategy

Table 3 shows how one of the best models of this approach was

Table 2

R^2 and errors (training) of the best models for both approaches (black-box - List_A and expert-in-the-loop - List_B), R^2 and errors of two examples taken from the literature*, and composition of all models (MDs). The lowercase letter in the model name indicates the specific MD combination of this cardinality for the best model.

Approach	Model name	#	MAE	RMSE	R^2
Black-box	ssA5_#21a_LR ¹	21	0.0091	0.0091	0.9686
	ssA2_#7a_LR ²	7	0.0159	0.0213	0.9143
Expert-in-the-loop	ssBPrev_#5a_LR ³	5	0.0173	0.0231	0.8987
	ssBPrev_#4a_LR ⁴	4	0.0189	0.0253	0.8791
	ssBPrev_#4m_LR ⁵	4	0.0195	0.0257	0.8748
	ssB6_#3a_RF ⁶	3	0.0177	0.0264	0.8694
	ssB4_#3b_RF ⁷	3	0.019	0.0264	0.8684
Literature*	Jabeen et al. [15] ⁸	4	0.0207	0.0289	0.8424
	Khan et al. [16] ⁹	6	0.0188	0.0244	0.8877
MDs	¹ Mi, nS, nR10, ChiA_X, ChiD/Dt, GATS3m, GATS4v, GATS1p, GATS8s, GGI5, GGI6, JGI3, SpMin1_Bh(s), SpMin3_Bh(s), nRCONHR, Si-111, SaasC, B02[C-Si], B06[C-N], F03[Br-Br], Hy ; ² Mi, nN, MAXDN, GATS2v, GATS1p, P_VSA_LogP_3, B05[C-Si] ; ³ Mi, GATS2p, MATS1p, ALOGP, ARR ; ⁴ Mi, GATS2p, MATS1p, ALOGP ; ⁵ Mi, GATS2p, MATS1p, ALOGP2 ; ⁶ GATS1e, SM1_Dz(p), nAB ; ⁷ Mi, GATS1p, GATS2p ; ⁸ nCsp2, ATSC2p, GATS1p, F01[C-F] ; ⁹ SpMaxA_D/Dt, Mi, MLFER_E, B01[O-Si], nCIC, SpMax_EA(bo)				

* QSPR models generated from the subsets of MDs used in models reported in the literature, which were trained with the same methods of the present work.

reached from the ssBPrev (cardinality 17). The process started by inferring QSPR models by removing each descriptor one by one, but always keeping the cardinality at 16. As a result, 17 models were obtained (ssBPrev_#16a to ssBPrev_#16q), whose R^2 was compared with the original model ssBPrev_LR ($R^2 = 0.9082$). When the value of R^2 decreased, as in the ssBPrev_#16a to ssBPrev_#16d models, these removed MDs were considered significant for the original model. This led to the detection of four relevant descriptors: Mi, GATS2p, MATS1p, and ALOGP, with which a new subset was created, and the corresponding models were inferred. In this way, the number of combinations of MDs analyzed was drastically reduced with respect to those evaluated during the backward elimination, which was processed automatically and carried out in the black-box approach. Table 3 shows that the ssBPrev_#4a_LR model has one of the best performances in this work ($R^2 = 0.8791$, also reported in Table 2). It should be clarified that by doing the same development with the other subsets, the same combination of descriptors was reached with the ssB2, obtaining the model ssB2_#4a_LR; the same happened with the ssB5, obtaining the model ssB5_#4a_LR, all equivalent (See Table S.4 of Supplementary material). This shows the relevance of the information contained in these four MDs, as will be seen in detail later on.

3.2.2. Visual analytics strategy

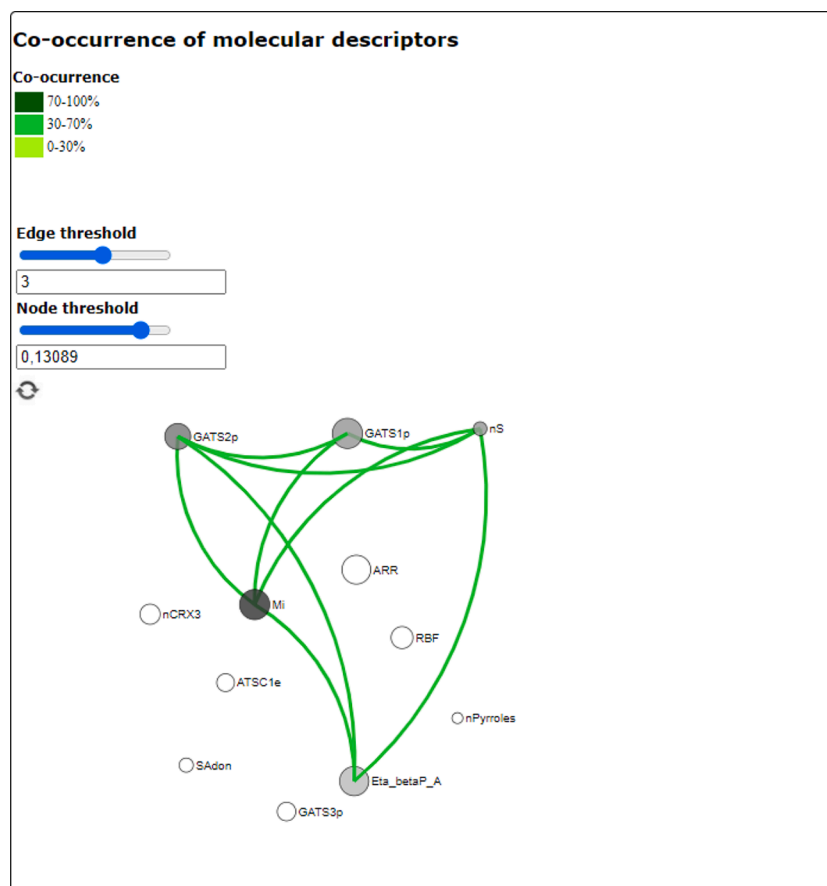
The intervention based on visual analytics is exemplified with ssB1. Fig. 2 shows a partial screenshot of the VIDEAN interface with ssB1 loaded. The 12 MDs appear as nodes, and the edges between them give information of appearance in pairs in more than two ssBs. This would indicate that the descriptors Mi, GATS1p, GATS2p, eta-betaP_A, and nS, whose edges are dark green, would be providing complementary information. For this reason, they were chosen, a new subset was formed, and the corresponding models were inferred (Table 4). Note that with a single step using visual analytics, a model with an $R^2 = 0.8793$ just below the original ssB1 (#12, $R^2 = 0.8913$) was obtained, but with a much lower cardinality (#5). This new model was taken as a reference for the next step. Each of its descriptors was eliminated (one by one), maintaining the cardinality at 4. It was found that the descriptor nS (number of Sulphur) did not contribute to accuracy because when it was removed, the value of R^2 practically did not change ($R^2 = 0.8740$). In this way, a last model of cardinality 4 was achieved, composed of Mi, GATS1p, GATS2p, and Eta_betaP_A, with an $R^2 = 0.8740$; it was very competitive in comparison with the original ssB1 ($R^2 = 0.8913$), especially considering that the new one, ssB1_#4a, has eight MDs less (Table 4). Fig. 3 shows the information space, which is an even more complete analysis. Four scatter plots are presented, one for each MD against RI; the behavior of each one is observed, occupying different zones of the information space. This leads us to think that each one contributes something different to the model. Kendall's correlation chart between the descriptors can be used as an additional analysis (Fig. 4). The correlation varies from 0.4 to 0.65, which can be interpreted as an intermediate level of linear correlation. Thus, from what is observed in Figs. 3 and 4, it can be concluded that these descriptors do not provide redundant information.

3.2.3. Physicochemical knowledge strategy

Finally, the expert chose among the best models of the strategies exemplified above, the best model with cardinality 4 from Table 2: ssBPrev_#4a_LR. This model was chosen not only for its performance but also because the combination of the information of its descriptors has very good complementation, as can be seen in Fig. 5 from the analysis with VIDEAN. Therefore, it is not surprising that this combination of descriptors was also found from the treatment of other subsets (ssB2 and ssB5), as mentioned above in the section Non-automatic elimination. However, the expert considered it necessary to add information about aromaticity, such as the ARR (aromatic ratio) MD; the results of the information spaces are shown in Figs. 5 and 6. It is evident that the behavior of this MD against RI provides valuable information, since the

Table 3Models inferred from ssBPrev with an expert's intervention using a non-automatic elimination method. R^2 values for training.

Model	#	Name	Machine learning methods (R^2)				
			RC	KNN	LR	RF	NN
ssBPrev	17	ssBPrev	0.8778	0.8453	0.9082	0.8830	0.8783
ssBPrev - GATS2p	16	ssBPrev_#16a	0.8630	0.8549	0.8768	0.8768	0.8692
ssBPrev - MATS1p	16	ssBPrev_#16b	0.8851	0.8457	0.8806	0.8906	0.8797
ssBPrev - ALOGP	16	ssBPrev_#16c	0.8727	0.8294	0.8892	0.8892	0.8677
ssBPrev - Mi	16	ssBPrev_#16d	0.8853	0.8431	0.8908	0.8906	0.8707
ssBPrev - ATSC2e	16	ssBPrev_#16e	0.8770	0.8438	0.9050	0.8881	0.8883
ssBPrev - GATS1e	16	ssBPrev_#16f	0.8712	0.8334	0.9069	0.8857	0.8791
ssBPrev - N%	16	ssBPrev_#16g	0.8748	0.8538	0.9069	0.8876	0.8705
ssBPrev - nAB	16	ssBPrev_#16h	0.8714	0.8403	0.9074	0.8876	0.8819
ssBPrev - SpMax1_Bh(p)	16	ssBPrev_#16i	0.8649	0.8352	0.9074	0.8862	0.8560
ssBPrev - F04 (C-Cl)	16	ssBPrev_#16j	0.8864	0.8460	0.9078	0.8872	0.8844
ssBPrev - nCl	16	ssBPrev_#16k	0.8827	0.8457	0.9088	0.8883	0.8840
ssBPrev - GATS1p	16	ssBPrev_#16l	0.8793	0.8471	0.9090	0.8797	0.8887
ssBPrev - SM1-Dz(p)	16	ssBPrev_#16m	0.8521	0.8457	0.9090	0.8830	0.8783
ssBPrev - Eta_betaP_A	16	ssBPrev_#16n	0.8770	0.8601	0.9092	0.8879	0.8832
ssBPrev - nCconj	16	ssBPrev_#16o	0.8804	0.8473	0.9094	0.8881	0.8819
ssBPrev - nF	16	ssBPrev_#16p	0.8599	0.8453	0.9095	0.8199	0.8864
ssBPrev - nS	16	ssBPrev_#16q	0.8823	0.8470	0.9105	0.8877	0.8795
ssBPrev#4a	4	ssBPrev_#4a	0.8084	0.8261	0.8791	0.8470	0.8653
ssBPrev#4a MDs	Mi GATS2p MATS1p ALOGP						
ssBPrev MDs	Mi GATS2p F04[C-Cl] GATS1e GATS1p N% nS ALOGP ATSC2e Eta_betaP_A MATS1p nAB nCconj nCl nF SM1-Dz(p) SpMax1_Bh(p)						

**Fig. 2.** Screenshot of the VIDEAN tool showing the pairwise co-occurrences (links) graph view of 12 MDs (nodes) of ssB1. Threshold is fixed in co-occurrence in three models (all pairs of descriptors that simultaneously appear in more than two models are linked).

R^2 increases from 0.8791 to 0.8987. This could be due to the fact that values equal to zero are reported in molecules that do not have aromaticity, and this would be marking a difference with the molecules that do have it. This proves that aromaticity is related to the property, as the π electrons intervene in the movements mentioned in Section 2.1. This new model was called ssPrev_#5a_LR (Table 2).

Note that during both strategies, the non-automatic elimination and visual analytics, the same relevant MDs were generally reached but in a much shorter time frame when the latter strategy was used. Although *a priori* the models ssBPrev_#4a_LR and ssBPrev_#5a_LR seemed to be the best (Table 2), the external validation had to be carried out and the performance parameters analyzed to make the definitive choice of the

Table 4

Models inferred from ssB1 with an expert's intervention using visual analytics. The R^2 values for model training (the maximum values are highlighted in bold).

Model	#	Name	Machine learning methods (R^2)				
			RC	KNN	LR	RF	NN
ssB1	12	ssB1	0.8913	0.8739	0.8834	0.8902	0.8285
Mi GATS1p GATS2p eta-betaP_A nS	5	ssB1_#5a	0.8512	0.8451	0.8793	0.8675	0.8363
Mi GATS1p GATS2p eta-betaP_A	4	ssB1_#4a	0.8597	0.8274	0.8740	0.8699	0.8455

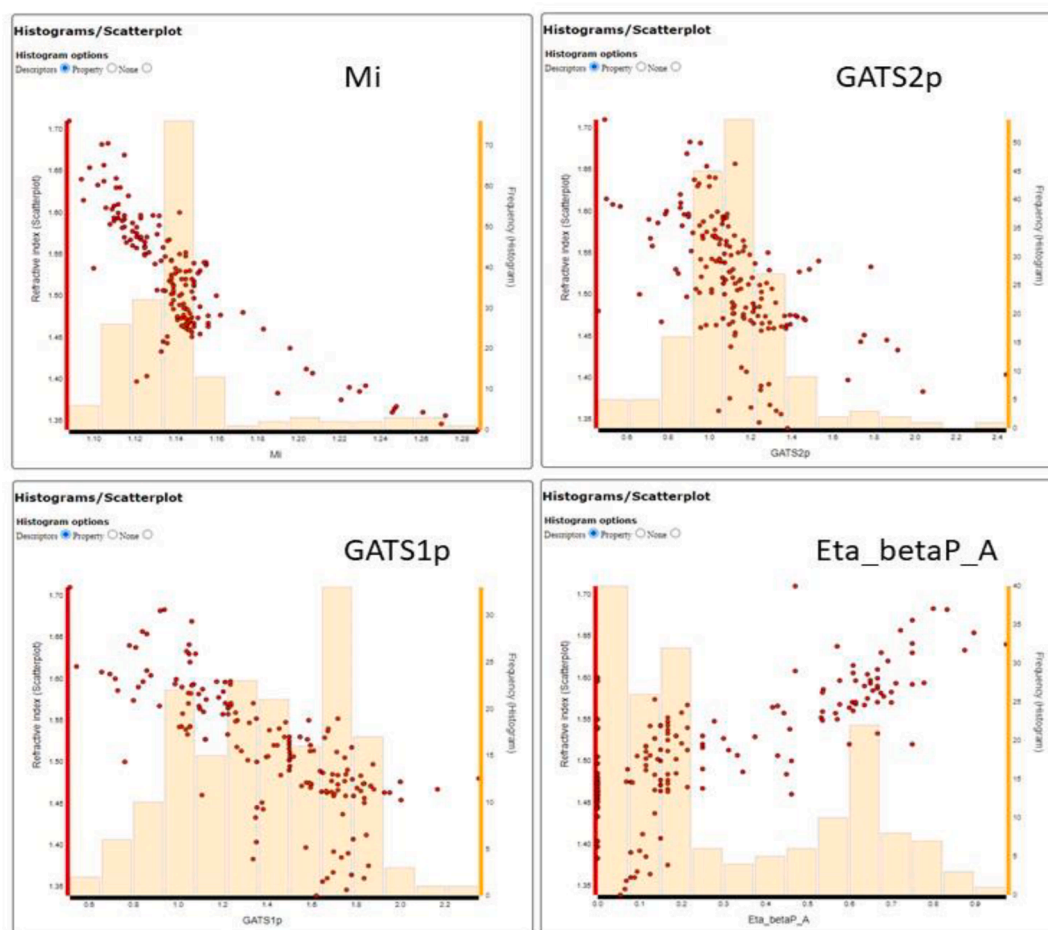


Fig. 3. Histograms and scatter plots of four MDs (Mi, GATS1p, GATS2p, and Eta_betaP_A) vs. the RI property.

best model for this work. Therefore, in the following section, the corresponding external validation of the best models of both approaches is carried out and also compared with two models taken from the literature.

3.3. External validation of QSPR models

Table S.5, in Supplementary Material, presents the external validation results of the best models of both the black-box approach and the expert-in-the-loop approach, together with the results of the models introduced by Jabeen et al. and Khan et al. The performances of the four best models in our work are similar in terms of MAE, RMSE, and R^2 , but differ greatly in cardinality. In particular, the model with the highest number of descriptors (ssA5_#21a_LR) is the only one that presents a remarkable difference between its training and the external validation values corresponding to MAE and RMSE (gap). Therefore, the appreciable increase of these values during the external validation would indicate a possible problem of model overfitting, which is to be expected in a QSPR model integrated by such a high number of MDs. With respect

to the model ssA2_#7a_LR, it does not have a cardinality as high as the model of #21; therefore, the R^2 remains very similar between the training and the external validation (gap). The same is true for the ssBPrev_#4a_LR and ssBPrev_#5a_LR models.

We also decided to compare the outcome of our work with some models in the literature that followed more traditional approaches. Note that the comparison was neither made with the models reported by Xu et al. [11,12] nor with the Duchowicz's model [14]. The reason was that we could not calculate the descriptors used by them, since our computational representations of the materials not available in our DB were required. Therefore, we only considered two reports, Jabeen et al. and Khan et al. [15,16], whose sets of descriptors we used to infer new models executed using the WEKA tool and our DB. We used the same ML methods and parameterizations that had been applied during the model inference stage (Section 2.4, Black-box approach) to make a fair comparison. With respect to the three tests to analyze the external validation proposed in Section 2.5, the six models in Table S.5 (four of our own and two from the literature) passed the first test as none of them obtained a gap >0.05. As for the second test, the hypothesis test, for the model

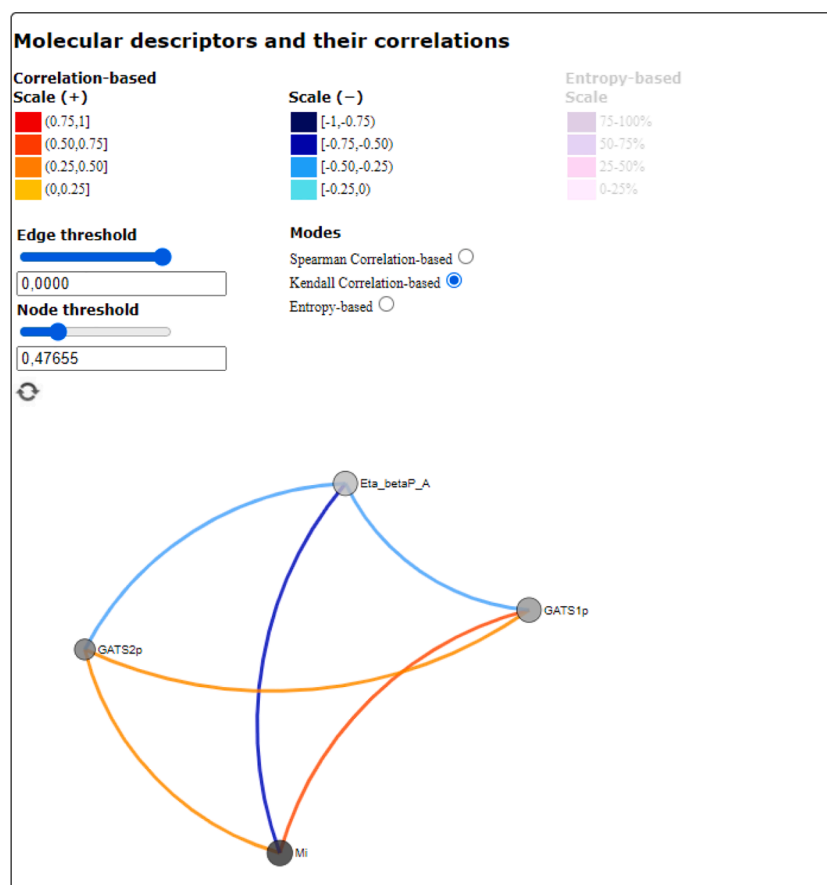


Fig. 4. Screenshot of the VIDEAN tool showing Kendall's correlation graph view for four descriptors (nodes): Mi, GATS1p, GATS2p, and Eta_betaP_A. The correlation values between each pair of descriptors are represented in a red range for positive values and in a blue range for the negative ones.

called ssA5_#21a_LR, the H0 could not be rejected; that is, it did not pass the test and therefore was not taken into account for the next one. With respect to the third test, the remaining five models generated quality parameters corresponding to “Good” predictions according to the algorithm presented by Roy *et al.* [36].

The performances of the models ssA2_#7a_LR and ssBPrev_#5a_LR are very similar ($R^2 = 0.9143$ and $R^2 = 0.8987$ respectively), but the difference is that the cardinality of the model intervened by the expert has ~30% less MDs. With regard to the two models generated by the expert's intervention (#4 and #5), both are competitive, presenting an R^2 of 0.8793 and 0.9107, respectively, for external validation. However, model with #5 has the most complete information on the composition of MDs (Figs. 5 and 6), as discussed at the end of Section 3.2. It is composed of: Mi (mean first ionization potential_scaled on Carbon atom), GATS1p (Geary autocorrelation of lag 1 weighted by polarizability), GATS2p (Geary autocorrelation of lag 2 weighted by polarizability), ALOGP (Ghose-Crippen octanol–water partition coeff. logP), and ARR (aromatic ratio). As discussed throughout this paper, these MDs manage to capture information related to the different aspects of polarity/polarizability. Because of all the discussions so far, it was decided to choose ssBPrev_#5a_LR as the best model for this work. Thus, when the expert intervenes, especially using visual analytics for feature engineering, low cardinality models can be obtained but with high contribution of physicochemical information related to the property, in a relatively shorter time with respect to the strategy without visual analytics.

With regard to the models reported in the literature by Jabeen *et al.* [15] and Khan *et al.* [16], both have R^2 and cardinalities similar to ours (Table S.5). The model reported by Jabeen *et al.* has cardinality 4, an externally validated R^2 of 0.8578, whereas the model by Khan *et al.* has cardinality 6 and an externally validated R^2 of 0.9185 (Table S.5). Note

that the four MDs in the model by Jabeen *et al.* are directly related to the aspects that determine polarity/polarizability in the molecule, namely nCsp2 (number of sp2 hybridized Carbon atoms), ATSC2p (Centred Broto-Moreau autocorrelation of lag 2 weighted by polarizability), GATS1p (Geary autocorrelation of lag 1 weighted by polarizability), and F01 [C-F] (Frequency of C - F at topological distance 1). Perhaps F01 [C-F] does not seem to be directly related *a priori*, but because Fluorine is a very electronegative atom, a molecule containing it has a marked difference in polarity from one without Fluorine. The composition of the model proposed by Khan *et al.* [16] also has this relationship. That is, SpMaxA_D/Dt (normalized leading eigenvalue from distance/detour matrix), Mi (mean first ionization potential_scaled on Carbon atom), MLFER_E (from PaDEL [39]: molecular linear free energy relation), B01 [O-Si] (Presence/absence of O-Si at topological distance 1), nCIC (number of rings_cyclomatic number), and SpMax_EA(bo) (leading eigenvalue from edge adjacency mat. weighted by bond order) are all related with polarity aspects. Fig. 7 shows that some MDs appear in several models, being Mi the most frequent, since it appears in five out of the six models in Table S.5, followed by GATS1p that appears in three of them. Note that these two MDs appear in our models and also in the model by Khan *et al.*, in spite of having followed different methodologies, which proves once again that these MDs have a preponderant relationship with this property.

3.3.1. Random experiments

Finally, once the best model in the present work was selected and passed the external validation test, it only remained to discard the possibility of a correlation by chance. Fig. 8 shows the results obtained in terms of mean R^2 and its standard deviation when running a Y-Randomization experiment 1000 times for our best model and the two

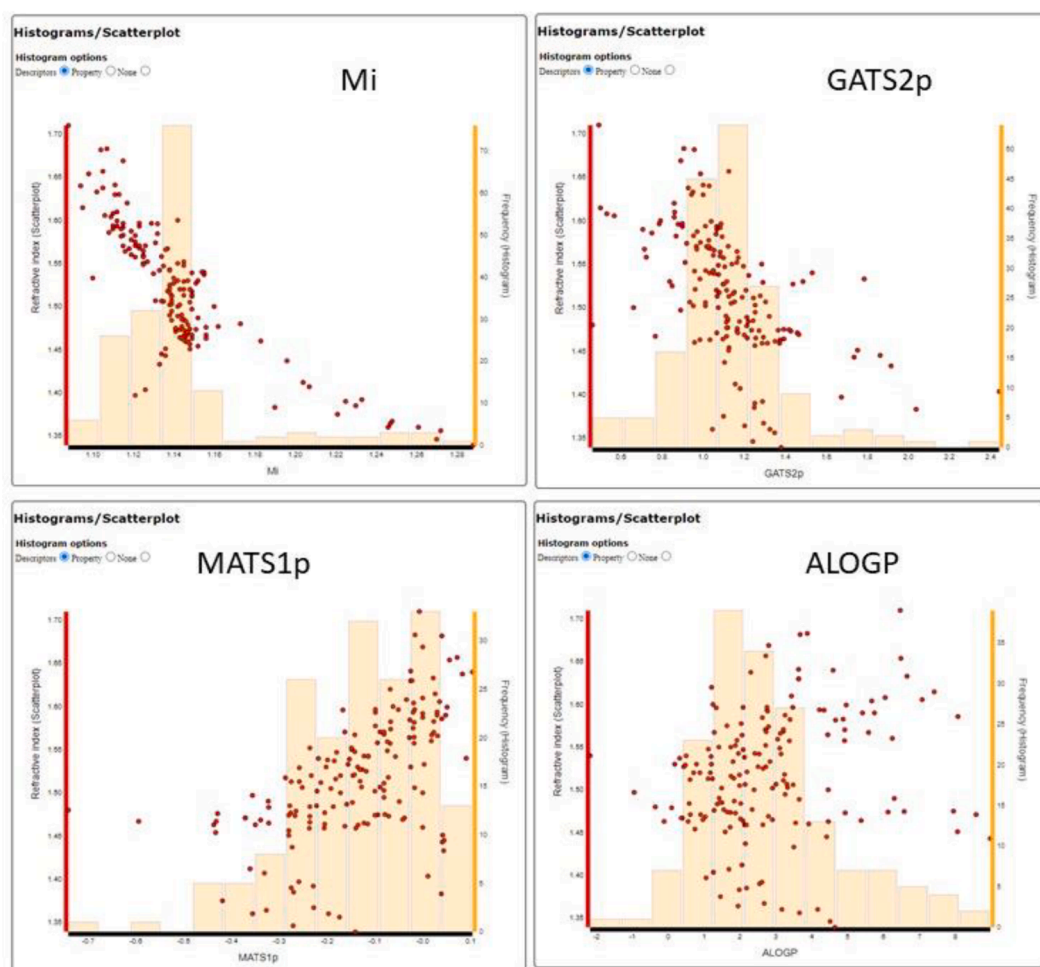


Fig. 5. Histograms and scatter plots of four MDs (Mi, GATS1p, GATS2p, and ALOGP) against the RI property.

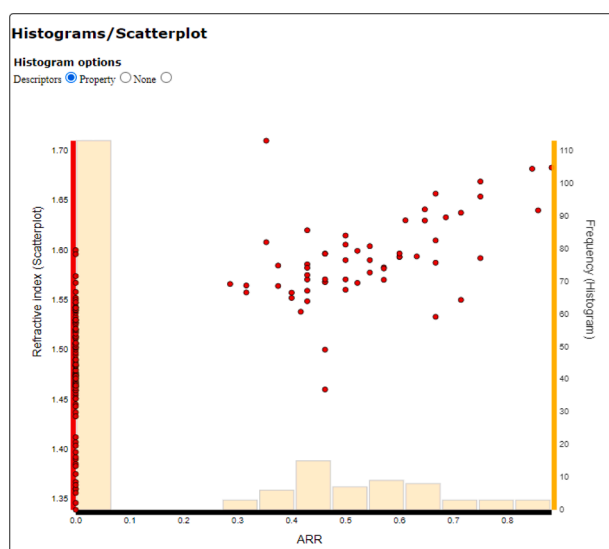


Fig. 6. Histogram and scatter plot of the ARR MD against the RI property.

models reported in the literature. The three of them successfully passed this test, since they have an R^2 (light blue) well above the one achieved with random experiments (yellow).

4. Conclusions

In the present work, QSPR techniques were used to infer prediction models of the RI of polymers, introducing an expert-in-the-loop approach with the aim of enriching black-box models with physico-chemical knowledge. This optical property is of special interest, since very specific values are required according to the industrial application (plastic optical fibers, anti-reflective coatings, and lenses, etc.). Therefore, the demand for new low-cost materials that meet these market needs has increased exponentially.

Data curation is a key step in QSPR modeling; for this reason, special care was taken in setting up the DB, since some inconsistencies and errors were found in the data extracted from the literature. We worked using ML with a novel expert-in-the-loop approach and compared it with a classical black-box approach. The expert made decisions from the beginning of the process proposing to reduce the DB descriptors to only those that somehow contemplate aspects related to polarity/polarizability. Later, in the inference stage of the models, the expert applied two interaction strategies to reduce the model cardinality, one following a non-automatic elimination process and another based on visual analytics, obtaining good results much faster with the latter. In addition, the enrichment of the models through the expert's knowledge was present in each methodological stage, except in the first automatic FS stage necessary to reduce the dimension and establish a number of human-treatable characteristics. As a result, a model was obtained to estimate the RI property of polymers called ssBPrev_#5a_LR ($R^2 = 0.8987$), competitive with others in the literature. Moreover, it could be verified

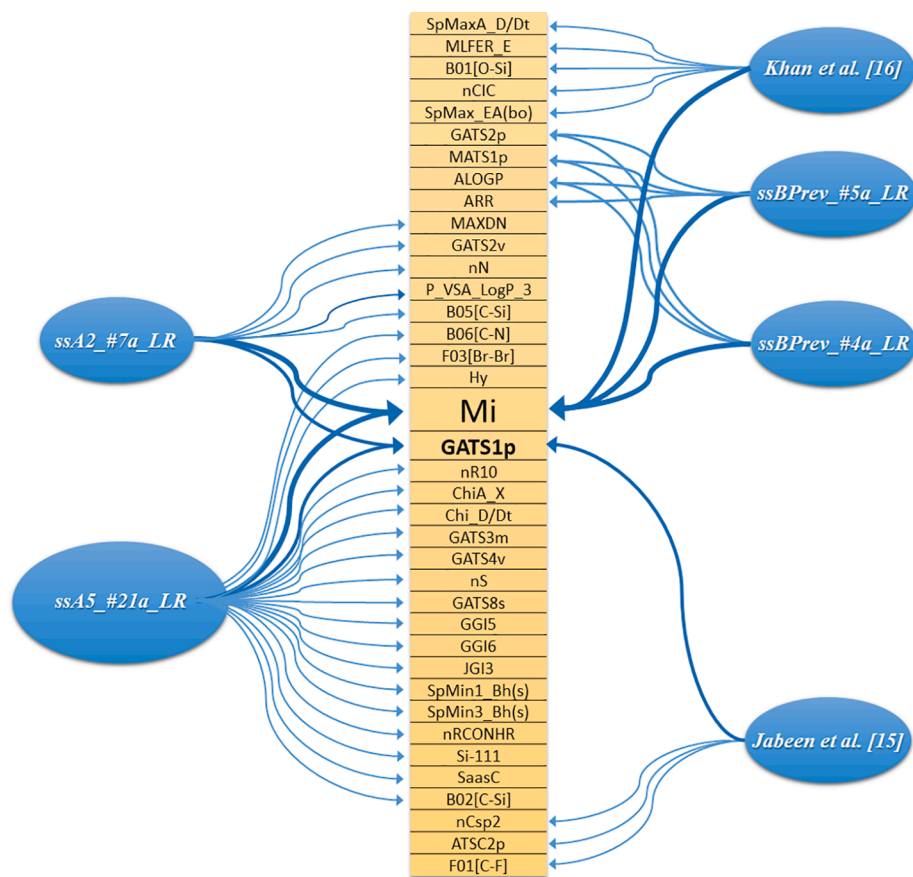


Fig. 7. MDs shared by our best models and two models from the literature: Jabeen et al. [15] and Khan et al. [16].

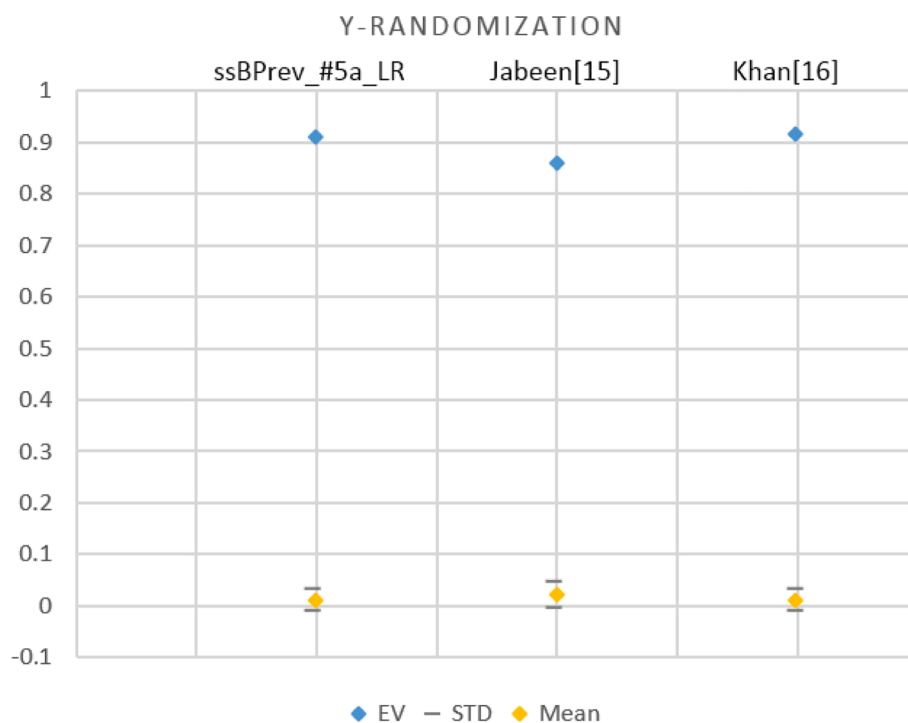


Fig. 8. Results for the mean R^2 and standard deviation (STD) of the 1000 Y-Randomization trials corresponding to our best model and the models proposed by Jabeen et al. [15] and Khan et al. [16], contrasted with External Validation (EV) of the original models.

by visual analytics that its five descriptors covered the information space, and it was also possible to establish the physicochemical relationships of each of them with RI. In this way, we consider that the obtained model achieves the desired robustness, supporting the hypothesis explored in this work: When experts intervene by providing knowledge in the different modeling stages, they can effectively contribute to reduce the cardinality and improve the interpretability of QSPR models while preserving their levels of accuracy.

The main goal of our research project is to provide *in silico* testing tools to designers of new polymeric materials to significantly shorten their development time. All efforts to add reliability and interpretability to QSPR models are worthwhile, as they are intended to support and assist designers of new materials in the early stages, prior to synthesis. In this sense, a model that was created from its inception using IML with expert-in-the-loop ensures that no random (by chance) features were entered into the model. Besides, as the model is based on the knowledge of the structure–property relationship, it can be applied with confidence to unseen molecules, provided that they are within the domain of applicability.

5. Data availability

The raw data required to reproduce these findings are available to download from Appendix A. Supplementary data.

Author contributions

M.F.D. and I.P. conceived the study and drafted the general structure and organization of the paper. They also provided funding acquisition.

S.A.S, F.C., and M.F.D. carried out the curation and debugging of the database.

S.A.S, F.C., and M.F.D. conducted and ran all the experiments for the different phases of the QSPR modeling and drafted the discussion on the importance of statistical parameters.

S.A.S, F.C., and M.F.D. performed the visual analytic study using VIDEAN software and drafted the discussion about relationships among the molecular descriptors selected for the QSPR models.

M.F.D. made the physicochemical interpretation and drafted the discussion of QSPR models.

All the authors analyzed the experiment results and collaborated in the writing of different sections of the manuscript. They revised and approved the submitted manuscript.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.commatsci.2021.110460>.

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